

VINTON VETERANS MEMORIAL, IA, USA

WMO: 720326

Lat: 42.219N

Lon: 92.026W

Elev: 258

StdP: 98.27

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04981

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.4	(c) -18.9	(d) -25.7	(e) 0.4	(f) -21.5	(g) -22.9	(h) 0.5	(i) -18.5	(j) 12.2	(k) -4.6	(l) 11.0	(m) -6.1	(n) 2.7	(o) 300	(p) 0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	33.0	23.8	32.4	23.5	30.6	22.5	25.7	31.6	24.7	29.9	23.6	28.0	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.6	28.6	22.9	18.2	27.2	22.3	17.5	26.6	80.6	31.1	76.0	30.1	71.8	28.0	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.8	(b) 9.1	(c) 8.2	(e) -26.6	(f) 35.9	(g) 3.1	(h) 1.2	(i) -28.9	(j) 36.7	(k) -30.7	(l) 37.4	(m) -32.4	(n) 38.1	(o) -34.7	(p) 38.9		
(a) 10.8	(b) 9.1	(c) 8.2	(e) -26.9	(f) 27.7	(g) 2.9	(h) 1.0	(i) -29.0	(j) 28.4	(k) -30.7	(l) 29.0	(m) -32.3	(n) 29.6	(o) -34.4	(p) 30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-6.0	-4.2	3.0	9.7	16.5	21.8	23.4	21.9	18.4	11.0	3.6	-3.0
	DBStd	11.70	6.91	7.02	6.70	5.46	5.04	3.34	3.33	2.84	4.74	5.31	5.76	6.39
	HDD10.0	1867	496	400	234	70	6	0	0	0	2	51	204	404
	HDD18.3	3627	754	632	476	262	96	7	2	3	55	234	443	662
	CDD10.0	1779	0	1	17	62	209	355	416	370	254	83	11	1
	CDD18.3	498	0	0	1	4	41	112	160	115	57	8	0	0
	CDH23.3	4768	0	0	9	59	416	1067	1598	986	554	78	1	0
	CDH26.7	1589	0	0	1	8	127	366	608	280	186	14	0	0
Wind	WSAvg	3.6	4.1	4.1	4.3	4.7	4.0	3.2	2.5	2.3	2.8	3.5	4.1	3.9
Precipitation	PrecAvg	937	31	29	56	90	123	144	122	111	82	70	62	29
	PrecMax	1466	63	65	127	194	235	270	325	292	132	168	158	57
	PrecMin	783	11	4	4	34	21	39	17	26	26	12	8	6
	PrecStd	167	13	16	38	41	47	60	77	63	28	45	42	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	16.4	24.9	27.7	33.0	35.3	35.6	32.9	34.4	28.0	21.9	14.3
		MCWB	6.1	10.0	16.3	15.8	21.9	24.4	25.5	24.3	23.5	18.8	14.3	12.5
	2%	DB	6.1	10.8	20.1	25.1	29.9	32.7	33.7	32.2	32.2	25.5	17.4	10.4
		MCWB	3.9	7.1	13.4	15.1	19.6	22.8	25.0	24.2	22.6	16.8	11.9	8.1
	5%	DB	4.0	7.0	16.1	22.3	27.4	31.3	32.5	29.7	28.6	22.8	14.8	7.4
		MCWB	2.1	4.1	11.6	13.4	17.9	22.0	24.2	22.8	20.6	16.3	10.4	5.4
	10%	DB	2.4	4.8	12.5	19.7	25.4	28.0	30.2	27.7	26.3	20.6	12.4	4.7
		MCWB	1.3	2.5	8.6	12.4	17.5	20.3	22.8	21.4	19.1	15.4	8.6	3.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	11.7	17.1	18.1	22.9	26.5	27.6	26.9	25.0	21.9	15.2	11.9
		MCDB	9.2	16.7	23.5	24.6	31.3	32.8	33.6	31.3	31.8	25.0	19.1	12.7
	2%	WB	4.3	7.4	14.7	16.6	21.5	24.6	26.2	25.2	23.6	19.9	13.2	8.3
		MCDB	6.5	10.7	18.6	23.1	27.9	30.2	32.2	29.8	28.0	23.6	17.2	9.7
	5%	WB	2.5	4.5	11.8	15.0	20.2	23.3	25.0	23.8	22.4	17.6	10.9	5.2
		MCDB	3.6	6.8	14.9	20.7	25.2	28.3	30.9	27.7	26.7	21.2	14.4	7.2
	10%	WB	1.4	2.3	8.9	13.0	19.0	22.1	23.9	22.8	21.2	15.2	8.8	2.9
		MCDB	2.4	4.0	12.4	18.5	23.6	26.9	28.4	26.3	25.0	19.7	11.9	4.1
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.1	9.6	11.5	11.9	11.3	11.5	11.4	12.5	11.3	9.7	8.1
		MCDBR	10.0	12.1	14.3	16.3	14.1	14.1	12.9	12.8	14.1	14.3	13.6	10.5
		MCWBR	8.1	8.8	8.8	8.9	6.3	6.4	5.9	6.4	6.2	7.7	9.1	8.5
	5% WB	MCDBR	8.8	11.9	12.7	13.4	12.1	11.7	11.9	10.8	11.3	11.3	13.1	10.0
MCWBR		7.4	8.8	8.3	8.5	6.5	6.3	6.1	6.2	5.9	8.0	9.7	8.3	
Clear-Sky Solar Irradiance	taub	0.284	0.308	0.331	0.377	0.409	0.422	0.435	0.415	0.385	0.338	0.299	0.282	
	taud	2.387	2.329	2.382	2.278	2.257	2.267	2.214	2.279	2.324	2.455	2.488	2.423	
	Ebn at Noon	863	894	915	890	867	854	837	843	844	851	843	835	
	Edh at Noon	79	99	107	128	134	133	139	126	112	87	72	70	
All-Sky Solar Radiation	RadAvg	1.84	2.86	3.82	4.68	5.55	6.13	6.19	5.40	4.50	2.95	1.99	1.47	
	RadStd	0.19	0.25	0.38	0.39	0.39	0.55	0.40	0.34	0.34	0.36	0.16	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page